

NORAD

Strategic Game of Air Warfare

Computer Edition Rulebook

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Introduction

NORAD is a computer implementation of the 1977 Mishler Company board game, a simulation of a hypothetical nuclear air war between the United States and the Soviet Union in the early 1960s. NORAD is a two-sided game: one side is the American defender, the other is the Soviet (Russian) attacker.

NOTE: “Russian” and “Soviet” are used interchangeably throughout these rules, exactly as in the original game.

Either side, or both, may be played by a human at the same computer. The program can also play either side for you, using one of two built-in computer opponents. These rules explain how to choose an opponent (see Prepare to Play, below) but do not describe how it decides its moves.

Description of the Playing Pieces (Units)

Soviet units are red. The basic game includes real and decoy bombers; the optional rules add real and decoy sub-launched ballistic missiles.

American units are blue. They include real and decoy fighters and real missiles. The Play Balance optional rule adds one hidden American missile decoy. Canadian fighter units, marked with a “C,” appear only under the Canadian Air Defense optional rule.

Each bomber and fighter shows its movement factor next to its silhouette – the maximum number of squares it may move in a single turn. Every American, Canadian, and Soviet-missile-type unit is covered in the Appendix: Unit Reference at the end of this rulebook, with exact quantities and movement factors.

On screen, a real unit and its decoy counterpart look completely identical from the front – you cannot tell them apart by sight. A unit’s true identity is only revealed when it is destroyed, or when you hold TAB to peek at the backs of your own side’s units (never the opponent’s), or after the game ends, using “Reveal all units.”

Description of the Game Board

The game board covers most of North America. Russia, Siberia, and Cuba are represented as staging areas along the edges of the map, not as squares on it. Every city on the board shows its point value.

Use the mouse wheel to zoom in and out (centered on the cursor), and right-click and drag – or the arrow keys – to pan the map.

How to Win

The Soviet player wins by reaching 100 or more points, earned by bombing American cities.

The American player wins by preventing that from ever happening. The game ends in an American victory once every real Soviet attacker – bomber or sub-launched missile – has been destroyed, has already bombed its target (and so can never move again), is stranded with nothing left it could still reach, or (under the Assigned Targets optional rule) can never reach the one city it was assigned.

Prepare to Play

Choosing Players and Opponents

Before a game begins, the start screen asks you to choose:

- Game mode – Two players (hot seat): both sides are played by people taking turns at the same computer.
- Game mode – Solo, you play the American side: the computer plays the Soviet side.
- Game mode – Solo, you play the Soviet side: the computer plays the American side.
- AI Opponent (solo modes only) – Standard AI Opponent or Expert AI Opponent, for whichever side the computer is playing. The Expert opponent plays a stronger, more deliberate game; Standard is simpler and more predictable. A hot-seat game needs no opponent choice; a solo game cannot start until one is picked.
- Optional rules – a checklist of the seven rules described under NORAD Optional Rules, below. All default ON except Assigned Targets.
- Game Options – Classic Movement (see How to Move, below).

American Player

The American player places ALL American units on city squares before the game begins. Units are placed face-up (silhouette showing).

There is no limit on how many real or decoy FIGHTERS may share a single city, but no more than ONE missile – real, or the Play Balance decoy – may be placed per city.

NOTE: If Canadian Air Defense is in effect, American units may only be placed in United States cities (Godthåb, though geographically in Greenland, counts as a US-eligible city, not a Canadian one); Canadian units may only be placed in Canadian cities.

Soviet Player

Depending on which optional rules are active, Soviet-side setup happens in this order, before movement begins:

1. Cuban-Based Placement (if enabled): stage up to five units of the fixed Cuban force on the row south of the board – before the American player sets up.
2. Assign SLBM Targets (if both Sub-Launched Missiles and Assigned Targets are enabled): assign a target coastal city to each real sub-launched missile – before the American player sets up.
3. American player sets up (see above).
4. Assign Bomber Targets (if Assigned Targets is enabled): assign a target city to each real bomber.
5. Movement turns begin (see How to Play, below).

The rest of the Soviet force – bombers, decoys, and, if enabled, sub-launched missiles – waits off-board and is staged into play turn by turn, as described next.

How to Play

The game proceeds in turns. The Soviet player always moves first each turn; the American player then has the option of moving any fighters.

Each Soviet turn begins with STAGING (unless the DEW Line has just broken – see Optional Rules): units are brought from the off-board pool onto the northern start line, and the Siberian start line if that optional rule is on, in preparation to enter play. At least FOUR units must be staged each turn – or however many remain, if fewer than four are left – until the entire Soviet force has been brought into play. Staged units sit one to a slot; they are never stacked while staged.

After staging, any staged unit may make its entry move onto the board (see How to Move, below). A unit does not have to enter the same turn it is staged.

Once the Soviet turn's movement and combat are complete, the American player may move any number of fighters, or none at all. Any fighter that is moved is expended for the rest of the game once its move is resolved – it either destroys an enemy unit or is simply removed if it found nothing there (see Combat, below).

Once a unit is placed on the board it can never leave, except by being destroyed. Destroyed units never return.

How to Move

Every square costs one step of a unit's movement allowance to enter. A unit is never required to use its full movement allowance.

Unit	Movement Allowance
Soviet bomber (real or decoy)	4 squares
American or Canadian fighter (real or decoy)	6 squares
Soviet sub-launched missile (optional)	1 square
American, Canadian, and Play-Balance decoy missiles	never move

There are two ways to move a unit on screen:

- Trace movement (the default): click each square in turn to trace out the unit's path; click the unit itself to stop it early. Click an already-moved unit again to undo and re-trace its move.
- Classic movement (a Game Option): click the destination directly and the unit jumps there along a computed path.

Entering the board: a staged Soviet unit's first move must land somewhere between the second and fourth row in from its start line – rows B through D from the northern or Siberian edge, or the equivalent rows after the DEW Line breaks (see Optional Rules).

NOTE: Aborting an entry move partway through returns the unit to the staged pool, as if it had never entered.

Soviet advance: after entering, a Soviet bomber or decoy must end each turn's move at least 2 rows further toward the southern edge than where it started that turn – unless it stops to bomb a city that turn, or has already reached row V, the southernmost row, where it remains for the rest of the game. Cuban-based units follow the reverse rule: they must advance at least 2 rows further NORTH each turn, unless they are bombing.

East/west limit: no unit may move more than 2 squares east or west in a single turn – unless the extra lateral distance ends that same move by bombing a city.

Units of either side may fly over squares occupied by enemy aircraft (real or decoy) without stopping or fighting – except that a Soviet unit must stop the instant it enters a square containing an American or Canadian missile. The American player then immediately decides whether to fire that missile (see Missile Combat, below).

Combat

Units and cities are eliminated by combat: fighters attacking Soviet units, missiles attacking Soviet units, and bombers attacking cities.

Fighter Combat

An American or Canadian fighter attacks only by moving onto the same square as a Soviet unit, during the American player's turn. Combat is resolved only after every fighter that is going to move this turn has finished moving. Both units are then revealed and removed – a fighter can never survive an engagement, whether the Soviet unit turns out to be real or a decoy.

If more than one fighter ends up on the same square, only one destroys the Soviet unit there; any extra fighters are also removed without effect.

NOTE: A fighter that moves to a square with no Soviet unit on it is simply removed – it found nothing to attack. A fighter may also fly through a square containing a Soviet unit without stopping, if it does not end its move there. Fighters cannot intercept a Soviet unit that enters their own city during the Soviet player's turn – only missiles can react on the Soviet player's turn.

Decoy fighters, American or Canadian, never attack; they are simply destroyed if the city they occupy is bombed.

Missile Combat

Any American or Canadian missile – real, or the Play-Balance decoy – may fire on a Soviet unit that stops in its city, the moment that unit enters. You choose whether to fire, destroying both units and revealing their identities, or hold fire, letting the Soviet unit continue moving or bomb the city that same turn.

A missile that holds its fire remains in place and challenges the next Soviet unit to enter that city the same way.

NOTE: A Soviet unit destroyed by a missile as it enters a city can never bomb that city. However, a different Soviet unit may still move into the same city later that turn and bomb it, if it is not also destroyed by a still-standing missile there.

Bombing Cities

A real Soviet bomber – or a real sub-launched missile, or the SLBM decoy simulating an attack – that stops in a city without being destroyed by a missile may bomb it. Bombing scores that city's point value for the Soviet player and destroys everything on that square, American and Soviet alike: any fighters or missiles stacked there are removed, and the Soviet player cannot check whether they were real or decoys until the game ends.

A bomber that bombs a city can never move again for the rest of the game. A city can only be bombed – and scored – once.

NOTE: Sub-launched missiles are a special case: a real missile must move its one square and bomb a coastal city on the turn immediately after it surfaces, or it is spent with no effect. See Russian Sub-Launched Ballistic Missiles, below.

NORAD - Optional Rules

Units and cities are eliminated by combat: fighters attacking Soviet units, missiles attacking Soviet units, The following optional rules may be turned on or off from the start screen, in any combination except where noted. All default ON except Assigned Targets.

The “DEW” Line

Anchorage and Godthåb represent the two keystone radar stations of the Distant Early Warning (“DEW”) Line.

If the Soviet player destroys BOTH cities, the DEW Line breaks: Soviet staging is suspended for the following two turns, and from the turn after that, new Soviet units stage directly on row H instead of the northern edge – reaching several rows further south the moment they enter play.

Separately, at any time before the DEW Line breaks, a decoy bomber staged from the northern or Siberian start line risks exposure the first time it reaches row H:

Anchorage / Godthåb Status	Chance a Crossing Decoy Is Exposed
Both still stand	50%
Exactly one still stands	25%
Both destroyed (DEW Line broken)	0%

NOTE: Each decoy is checked only once, the instant it first reaches row H. Real bombers and Cuban-staged decoys are never subject to this check.

Optional Russian Placement

Part I: Siberian-Based

Adds a second Soviet start line along the western edge of the board, in addition to the northern edge. Units may be staged from either line, in any combination.

Part II: Cuban-Based

Sets aside a fixed force of 3 real and 5 decoy bombers from the main Soviet pool. Before the American player sets up, the Soviet player may stage up to 5 of these 8 units, in any mix of real and decoy, on the row-V line at the southern edge. Any of the 8 left un-staged simply rejoin the main northern force instead of being wasted.

Cuban-based units may be brought into play, one at a time, on any of the Soviet player's turns, entering at the row-V squares just north of the staging line; their first move is limited to that same row. On subsequent turns they move under the reverse advance rule: at least 2 rows further NORTH each turn, unless they bomb a city that turn.

Russian Sub-Launched Ballistic Missiles

Adds 3 real and 1 decoy sub-launched missile. These may be brought into play on any Soviet turn, entering one square offshore, adjacent to any coastal city (no more than one missile per square). During the American player's following turn, these newly surfaced missiles may be attacked by fighters exactly as bombers are.

On the Soviet player's NEXT turn, a real missile must move its single square onto a coastal city and bomb it, or it is expended with no further effect; the decoy missile likewise moves one square to simulate an attack, but of course does no damage. While sitting in a city square, a sub-launched missile may be attacked by an American missile exactly as a bomber would be. Each sub-launched missile may attack at most one city.

NOTE: This rule requires Canadian Air Defense to also be enabled.

Canadian Air Defense

Adds 3 real and 1 decoy Canadian fighter, visually distinct from American fighters. These units may only be set up in Canadian cities; American units, in turn, may only be set up in United States cities (Godthåb is treated as a US-eligible city for this purpose, not a Canadian one). Canadian fighters and decoys are used exactly like American ones.

Assigned Targets

Before the game begins, the Soviet player assigns one target city to each real bomber, after the American player has set up, and, if Sub-Launched Missiles are also on, to each real sub-launched missile, before the American player sets up (this is the one exception – see Prepare to Play, above). A unit may only bomb the specific city assigned to it. If that city is destroyed by another unit, or otherwise becomes unreachable, the assigned unit can never score.

Play Balance

A rule unique to this computer edition, intended to help balance the two sides. It adds one American missile decoy – indistinguishable from a real missile from the front – to a coastal city, and removes one Soviet decoy bomber from the game, leaving 7 instead of 8.

The Computer Interface

This section describes the on-screen controls used to play NORAD. It does not restate the rules themselves – see the sections above for those.

Controls Reference

Action	Input
Select / trace a move / place a unit / press a button	Left click
End a traced move	Click the unit itself
Abort or restart a move	Click an already-moved unit, or press Esc mid-trace
Zoom	Mouse wheel
Pan the map	Right-click and drag, or the arrow keys
Peek at your own units' backs	Hold TAB
Hide all units (view the bare map)	Hold C
Open / close the Rules Summary	R
Cancel a selection / quit from Game Over	Esc

Movement: Trace Mode vs. Classic Mode

By default the game uses trace movement: click each square of the path you want a unit to take, in order, then click the unit itself to end its move there. Clicking a unit that has already moved this turn aborts that move and lets you re-trace it from scratch.

The Classic Movement game option instead lets you click a destination directly – the unit automatically plots a legal path and jumps there. Turn this on or off from the start screen before the game begins; it cannot be changed mid-game.

Resolving Combat and Bombing

Combat results and city bombings pause the game with a pop-up banner; click anywhere, or press Space or Enter, to continue. When American fighter combat is pending at the end of your movement turn, each affected square is revealed and resolved one at a time this way.

The In-Game Rules Summary

Press R at any time during your turn to bring up a full-screen summary of these rules – Objective, Setup order, Movement, Combat, and the Optional Rules currently in effect – along with a silhouette and current count of every unit type in your game. Press R again to close it.

Ending the Game

When the game ends, the final Soviet point total is shown. Press Esc to exit, or click “Reveal all units” to flip every remaining unit face-up and see which were real and which were decoys.

Appendix: City Point Values

Point values for every city on the game board, as implemented in this computer edition. Canadian cities are used only under the Canadian Air Defense optional rule; see Prepare to Play, above.

City	Point Value	Canadian
Anchorage	5	
Atlanta	5	
Birmingham	6	
Boston	5	
Chicago	9	
Cincinnati	6	
Dallas	5	
Denver	6	
Detroit	8	
Edmonton	5	Yes
Godthåb	5	
Houston	5	
Jacksonville	8	
Kansas City	6	
Key West	6	
Los Angeles	6	
Minneapolis	5	
Mobile	6	
New Orleans	6	
New York	9	
Norfolk	7	
Omaha	9	
Philadelphia	5	
Phoenix	5	
Pittsburgh	7	
Portland	5	
Quebec	5	Yes
San Antonio	5	
San Diego	8	
San Francisco	7	
Savannah	5	
Seattle	7	
St. Louis	5	
Toronto	6	Yes
Vancouver	5	Yes
Washington, D.C.	9	
Winnipeg	5	Yes

Appendix: Unit Reference

Standard NORAD Playing Pieces

Unit	Quantity	Movement Factor
Soviet Bomber (real)	23	4
Soviet Bomber Decoy	8 (7 under Play Balance)	4
American Fighter (real)	12	6
American Fighter Decoy	4 (5 under Play Balance)	6
American Missile (real)	5	never moves

Optional NORAD Playing Pieces

Unit	Quantity	Movement Factor
Soviet Sub-Launched Missile	3 real + 1 decoy	1
Canadian Fighter	3 real + 1 decoy	6
American Missile Decoy (Play Balance)	1	never moves